

WORK

Work is defined as the product of the applied nett force and its displacement in the direction of the nett force.

i.e.

$$W = Fs$$

where W = the work done (J-joules)
 F = the nett force applied to the object (N)
 s = the displacement of the object (m)

Work is a **scalar** quantity - no direction is required.

When work is done on an object, it changes the energy of the object.

One **JOULE** of work is done when the point of application of a net force of one newton moves an object through a distance of one metre in the direction of the force.

If the force is applied at an angle to the direction of movement, it is the **component of the force** responsible for the work done.

(This is Year 12 course work and will be covered then.)

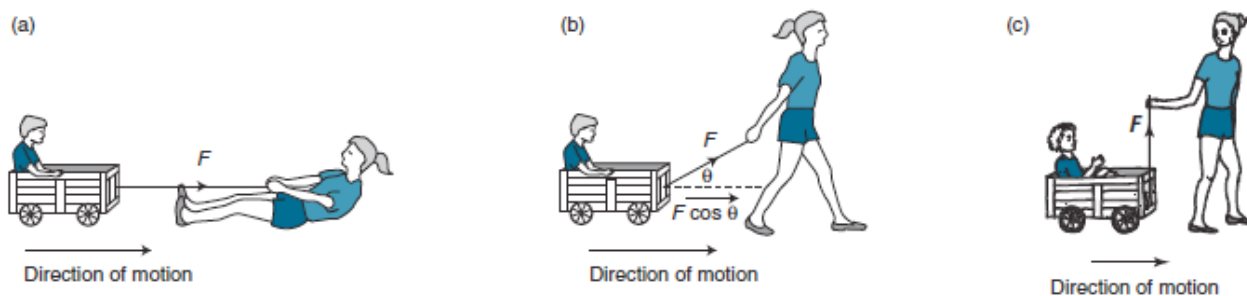


Figure 4.9

(a) If a force is applied in the direction of motion of the cart, then the force is at its most effective in moving the cart. (b) When the force is applied at an angle θ to the direction of motion of the cart, the force is less effective. The component of the force in the direction of the displacement, $F \cos \theta$, is used to calculate the work. (c) When the angle at which the force is acting is increased to a right angle ($\theta = 90^\circ$), then the component of the force in the direction of the intended displacement is zero and it does no work on the cart—provided of course that it doesn't lift the cart, in which case work would also be done against gravity.

If a force is applied to an object and it does not move, then no work has been done.

- e.g. 1. A person may push continuously against a parked car until exhausted, but if the car does not move, **no mechanical work has been done**.

However, the person has expended a significant amount of work internally (physiologically)!

2. Carrying a bag at a constant speed across the ground at a constant height has no work done on it.

Reason: the energy of the bag has not changed.
(See the section on Energy.)

Examples

Calculate the work done against gravity by an athlete of mass 60 kg competing in the Rialto Tower Run-up illustrated in Figure 4.10. Use $g = 9.8 \text{ m s}^{-2}$.

Solution

Only the weight force need be considered in this example as the work in the vertical direction is all that is required.

$$m = 60 \text{ kg}, g = 9.8 \text{ m s}^{-2}, s = \Delta h = 242 \text{ m}$$

$$\text{Force required} = \text{weight} = mg = 60 \times 9.8 = 588 \text{ N}$$

$$W = Fs$$

$$= 588 \times 242$$

$$= 142\,296 \text{ N m} = 1.4 \times 10^5 \text{ J}$$



Figure 4.10

The annual Rialto Tower Run-up is a race to climb 1254 steps up the Rialto Tower building in Melbourne. Contestants climb a vertical distance of approximately 240 m and the winner takes around 7 minutes to complete the race.

1.

2. The girl in Figure 4.9 pulls the cart by applying a force of 50 N at an angle of 30° to the horizontal. Assuming a force due to friction of 10 N is also acting on the wheels of the cart, calculate the work done if the cart is moved 10 m along the ground in a straight line.

Solution

$$F_1 = 50 \text{ N}, \theta = 30^\circ, F_f = 10 \text{ N}, s = 10 \text{ m}$$

$$\Sigma F = F_1 \times \cos 30^\circ - F_f$$

$$= 50 \times 0.866 - 10 = 33.3 \text{ N}$$

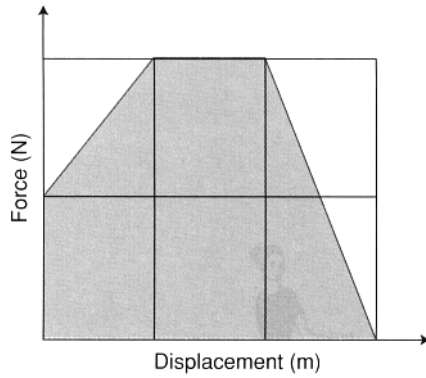
$$\text{Now } W = \Sigma Fs,$$

$$\text{and work} = 33.3 \times 10 = 330 \text{ J}$$

Force - Displacement Graphs

The area under such a graph gives a measure of the work done by the force.

This is particularly useful in situations where a force changes with displacement, such as in collisions.



$$\begin{aligned}\text{The area under the graph} &= W_{\text{done}} \\ &= F_{\text{ave}} s\end{aligned}$$

Thus it is possible to determine the average force which acted during the situation being studied.